MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 21: Tuesday 27 October 2026

(Week 11)

Remark. Q20.5 is leftover exercise from the Friday class in Week 10.

Remark. Q21.1 to Q21.5 (Spherical coordinates) are on Section 14.7 of Thomas' Calculus.



- Towards the end of the semester, you can get the exam dates, times, and venues and seat numbers for all your courses from Edurec:

Edurec $>$ Academics $>$ Examinations

Then click 'View Exam Timetable for Schedule'.

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Q20.5. A steel beam with a pointed end and a round end is modelled as the set of points in $\mathbb{R}^3$ in the cylinder of radius **1** centered at the origin, between the cone $z = x^2 + y^2 - 10$ and the hemisphere $z = \sqrt{1 - x^2 - y^2}$.

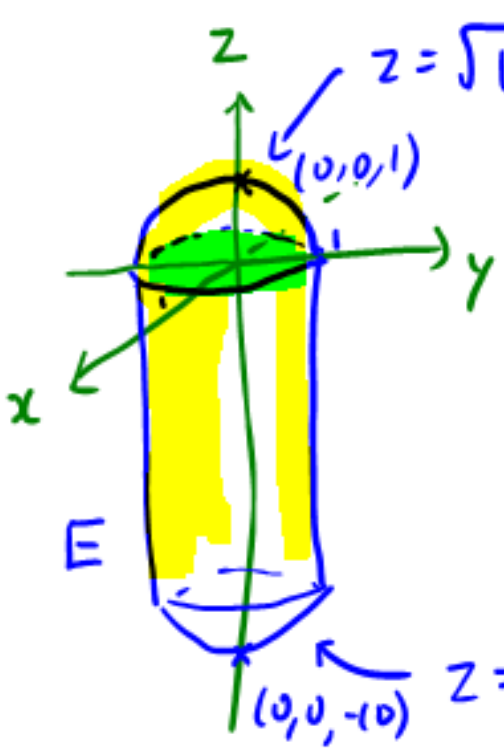
$x^2+y^2+z^2=1\ ,\ z\geq 0$

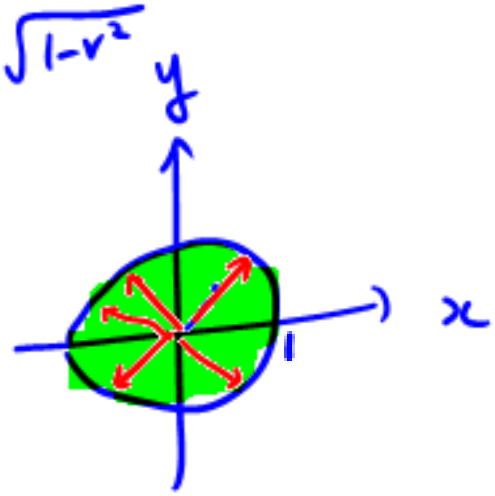
From the way that the steel beam is made, you deduce that its density $f(x, y, z)$ at a point (x, y, z) in the steel beam is proportional to the absolute value of its z-coordinate.

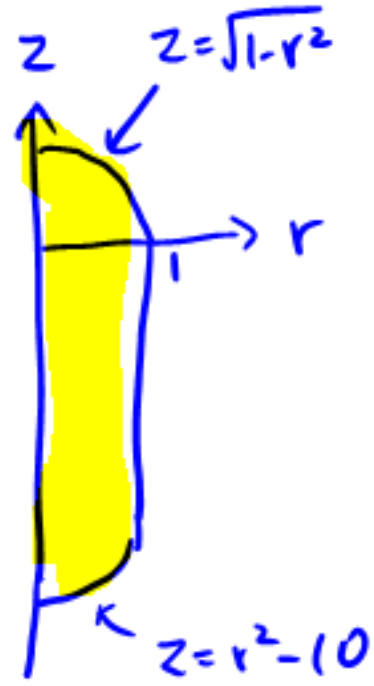
$f(x,y,z) = k|z|$

You also measured that the total mass of the steel beam is **5**.

Find a formula for the density function f.

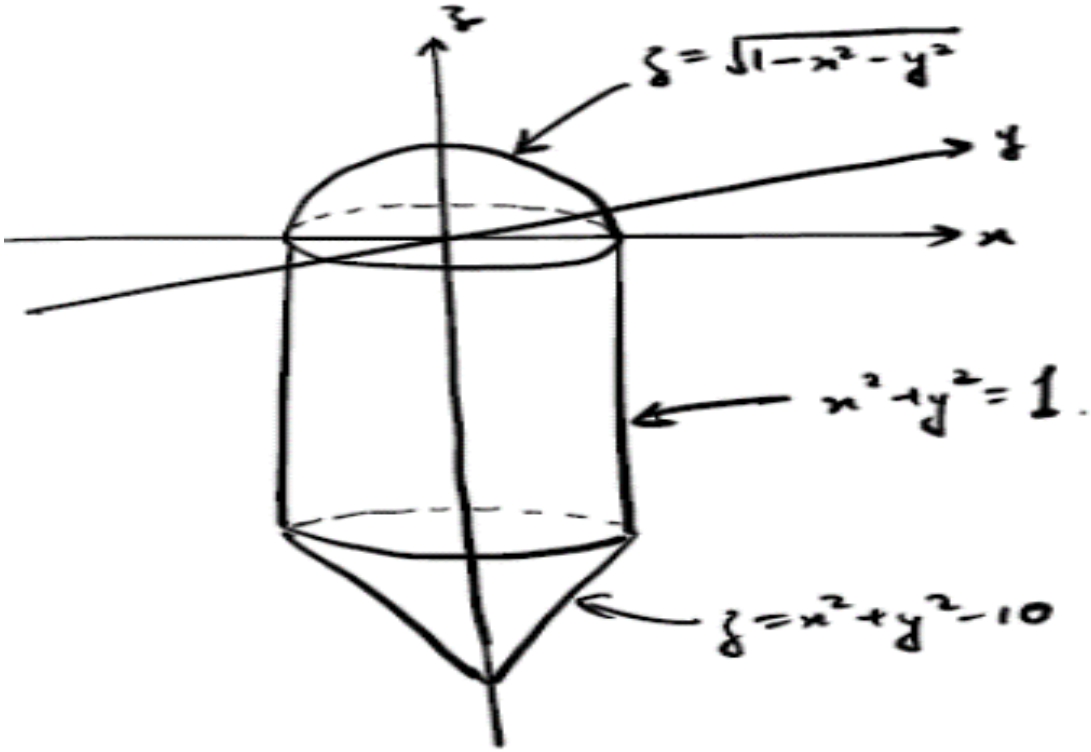






$$E = \{ (r,\theta,z) : \ 0 \le \theta \le 2\pi, \ 0 \le r \le 1, \ r^2-10 \le z \le \sqrt{1-r^2} \}$$

Solution:



$$|z| = \begin{cases} z & \text{if } z \geq 0 \\ -z & \text{if } z < 0 \end{cases}$$

Given that $f(x, y, z) = k|z|$ for some constant $k \in \mathbb{R}$.

Let

$$E = \{ (r, \theta, z) : 0 \leq r \leq 1, \ 0 \leq \theta \leq 2\pi, \ r^2 - 10 \leq z \leq \sqrt{1 - r^2} \}.$$

Then

density

$$5 = \iiint_E k|z|\, dV$$

$dV = r\, dz\, dr\, d\theta$

$$= \int_0^{2\pi} \int_0^1 \int_{r^2-10}^{\sqrt{1-r^2}} k|z|\, r\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^1 \left(\int_{r^2-10}^{0} -kz\, r\, dz \right) dr\, d\theta$$

$z \geq 0$

$z < 0$

$$+ \int_0^{2\pi} \int_0^1 \left(\int_0^{\sqrt{1-r^2}} kz\, r\, dz \right) dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^1 -kr \left[\frac{z^2}{2} \right]_{r^2-10}^{0} dr\, d\theta$$

$$+ \int_0^{2\pi} \int_0^1 kr \left[\frac{z^2}{2} \right]_0^{\sqrt{1-r^2}} dr\, d\theta$$

$$= \frac{k}{2}\int_0^{2\pi}\int_0^1 r(r^2-10)^2\,dr\,d\theta + \frac{k}{2}\int_0^{2\pi}\int_0^1 r(1-r^2)\,dr\,d\theta$$

$$= \frac{k}{2}\int_0^{2\pi}\int_0^1 \left(r^5-20r^3+100r+r-r^3\right)dr\,d\theta$$

$$= \frac{k}{2}\int_0^{2\pi}\left(\int_0^1 \left(r^5-21r^3+101r\right)dr\right)d\theta$$

$$= \frac{k}{2}\int_0^{2\pi}\left[\frac{r^6}{6}-\frac{21r^4}{4}+\frac{101r^2}{2}\right]_0^1 d\theta$$

$$= \frac{k}{2}\int_0^{2\pi}\left(\frac{1}{6}-\frac{21}{4}+\frac{101}{2}\right)d\theta$$

$$= \frac{k}{2}\cdot\frac{2-63+606}{12}\cdot\left[\theta\right]_0^{2\pi}$$

$$= \frac{k}{2}\cdot\frac{545}{12}\cdot 2\pi = \frac{545k\pi}{12}.$$

Hence we have

$$5 = \frac{545k\pi}{12} \implies k = \frac{12}{109\pi}.$$

Hence

$$f(x, y, z) = \frac{12}{109\pi}|z|.$$

cylindrical coordinates (r, θ, z)

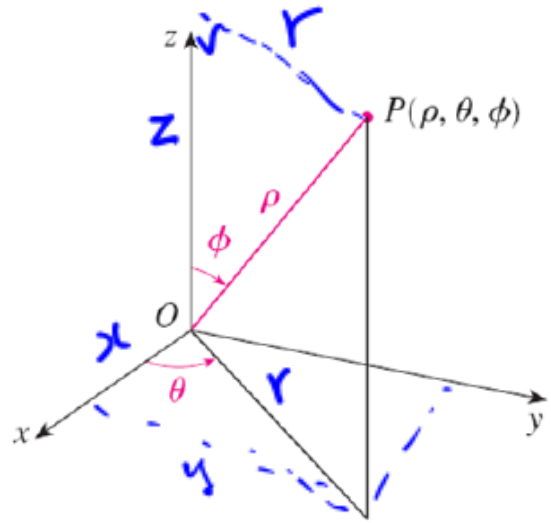
Triple integrals in spherical coordinates

Sometimes it is convenient to use **spherical coordinates** (ρ, ϕ, θ) to compute triple integrals.

Spherical coordinates (ρ, ϕ, θ):

- ρ is the distance from P to the origin O ($\rho \geq 0$).
- ϕ is the angle $\overrightarrow{OP}$ makes with the positive z-axis. ($0 \leq \phi \leq \pi$)
- θ is the same angle as in cylindrical coordinates.

$$\begin{cases} r = \rho \sin\phi \\ z = \rho \cos\phi \\ \theta = \theta \end{cases}$$



$$\rho = \sqrt{x^2+y^2+z^2} = \sqrt{r^2+z^2}$$

$$\begin{cases} x = \rho \sin\phi \cos\theta \\ y = \rho \sin\phi \cos\theta \\ z = \rho \cos\phi \end{cases}$$

Transformation between (x, y, z) and (ρ, ϕ, θ):

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi. \qquad (1)$$

Suppose R is a region in xyz-space, and G is the corresponding region in the $\rho\phi\theta$-space. Then

$$\iiint_R F(x, y, z)\, dV \qquad (2)$$

$$= \iiint_G F(\rho \sin\phi \cos\theta, \rho \sin\phi \sin\theta, \rho \cos\phi)\, \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$$

"$dV = \rho^2 \sin\phi \; d\rho\, d\phi\, d\theta$"

Derivation of (2): From the change of variables formula for triple integrals, we have

$$\begin{aligned}
&\iiint_R F(x,y,z)\,dV \\
&= \iiint_R f(x,y,z)\,dx\,dy\,dz \\
&= \iiint_G F(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi) \\
&\qquad \cdot \left|\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)}\right| d\rho\,d\phi\,d\theta.
\end{aligned}$$

Then (2) follows immediately from the following computation:

Recall from (1) that

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi.$$

So we have

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)}$$

$$= \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \phi} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \phi} & \frac{\partial y}{\partial \theta} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \phi} & \frac{\partial z}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \sin\phi \cos\theta & \rho \cos\phi \cos\theta & -\rho \sin\phi \sin\theta \\ \sin\phi \sin\theta & \rho \cos\phi \sin\theta & \rho \sin\phi \cos\theta \\ \cos\phi & -\rho \sin\phi & 0 \end{vmatrix}$$

$$
\begin{aligned}
&= \sin\phi\cos\theta \cdot \rho^2 \sin^2\phi \cos\theta \\
&\quad - \rho\cos\phi\cos\theta \cdot (-\rho\sin\phi\cos\phi\cos\theta) \\
&\quad + (-\rho\sin\phi\sin\theta) \cdot (-\rho\sin^2\phi\sin\theta - \rho\cos^2\phi\sin\theta) \\
&= \rho^2 \sin\phi\cos^2\theta(\sin^2\phi + \cos^2\phi) \\
&\quad + \rho^2 \sin\phi\sin^2\theta(\sin^2\phi + \cos^2\phi) \\
&= \rho^2 \sin\phi\cos^2\theta + \rho^2 \sin\phi\sin^2\theta \\
&= \rho^2 \sin\phi.
\end{aligned}
$$

$= \rho^2 \sin^3\phi \cos^2\theta$

$= \rho^2 \sin\phi \cos^2\phi \cos^2\theta$

$\sin^2\phi + \cos^2\phi = 1$

Q21.1. (a) Describe the surface S whose equation is given by $\rho = \sin\theta \sin\phi$ in spherical coordinates.

[Suggestion: First rewrite the equation in Cartesian coordinates.]

(b) Let D be the solid region bounded by S. Describe D in spherical coordinates.

Solution: (a) To draw the surface S, we first re-write the equation of the surface in terms of ~~Cartesian~~ rectangular coordinates (x, y, z).

Recall that the spherical coordinates (ρ, θ, ϕ) are related to the Cartesian coordinates (x, y, z) by

$$\begin{cases} x = \rho \sin\phi \cos\theta, \\ y = \rho \sin\phi \sin\theta, \\ z = \rho \cos\phi, \end{cases}$$

so that $\rho = \sqrt{x^2 + y^2 + z^2}$.

First we consider the case when $\rho \neq \mathbf{0}$.

$y = \rho \sin\phi \sin\theta$

$$\rho = \sin\phi \sin\theta \quad (1)$$

$$\Leftrightarrow \quad \rho^2 = \rho \sin\phi \sin\theta$$

$$\Leftrightarrow \quad x^2 + y^2 + z^2 = y$$

$$\Leftrightarrow \quad x^2 + y^2 - y + z^2 = 0$$

$$\Leftrightarrow \quad x^2 + \left(y^2 - y + \tfrac{1}{4}\right) + z^2 = \tfrac{1}{4}$$

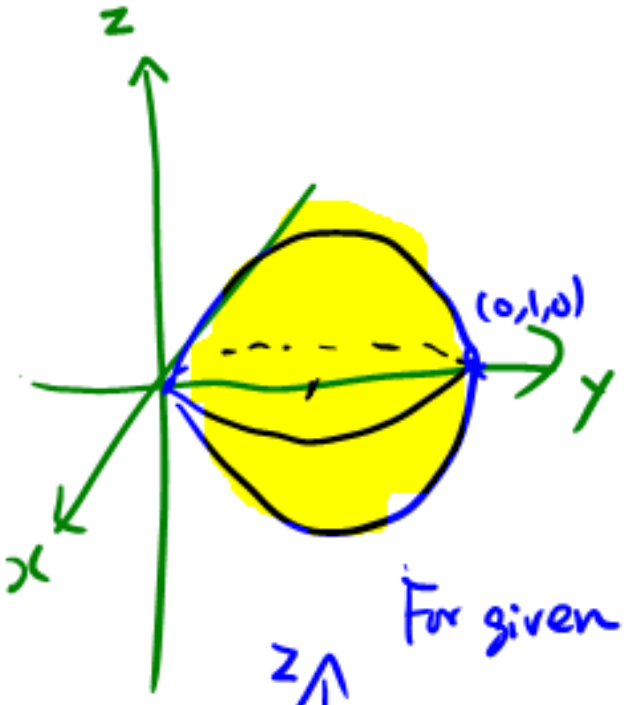
$$\Leftrightarrow \quad x^2 + \left(y - \tfrac{1}{2}\right)^2 + z^2 = \left(\tfrac{1}{2}\right)^2 \quad (2)$$

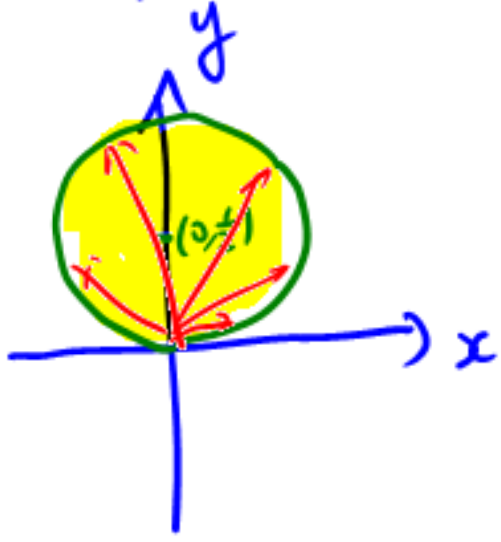
Sphere centered at $(0, \frac{1}{2}, 0)$ & of radius $\frac{1}{2}$.

$\rho = 0$

$(0,0,0)$ satisfies both (1) & (2)

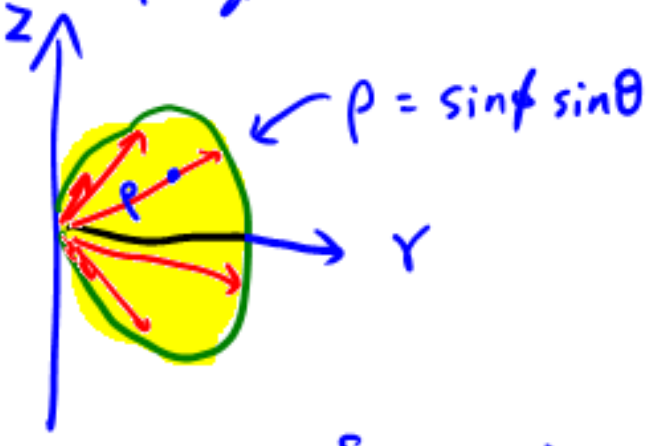
(b). D = solid region bounded by S





For given $0 \le \theta \le \pi$,

$\rho = \sqrt{r^2 + z^2}$



$0 \le \phi \le \pi$

$0 \le \rho \le \sin\phi \sin\theta$

$D = \{(\rho, \phi, \theta) : 0 \le \theta \le \pi,\ 0 \le \phi \le \pi,\ 0 \le \rho \le \sin\phi \sin\theta\}$

Note that $\rho = \mathbf{0}$ corresponds to the origin $\mathbf{(0, 0, 0)}$ which satisfies both (1) and (2).

Hence both equations (1) and (2) define the same surface $\boldsymbol{S}$.

From (2), we know that $\boldsymbol{S}$ is the sphere of radius $\dfrac{\mathbf{1}}{\mathbf{2}}$ and centered at $\mathbf{(0, \dfrac{1}{2}, 0)}$.

Q21.2. Evaluate

$$\iiint_W e^{(x^2+y^2+z^2)^{3/2}}\, dx\, dy\, dz,$$

where W is the unit ball centered at the origin, i.e.,
$W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$.

Solution: Rewriting W in spherical coordinates, we have

$$W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$$

$$= \{(\rho, \phi, \theta) : 0 \leq \phi \leq \pi, \ 0 \leq \theta \leq 2\pi, \ 0 \leq \rho \leq 1\}$$

$$\iiint_W e^{(x^2+y^2+z^2)^{3/2}}\, dV$$

$$= \int_0^{2\pi}\int_0^{\pi}\left(\int_0^1 e^{(\rho^2)^{3/2}}\cdot \rho^2 \sin\phi\, d\rho\right) d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\phi\left(\int_0^1 e^{\rho^3}\cdot \rho^2\, d\rho\right) d\phi\, d\theta$$

$$= \left(\int_0^1 e^{\rho^3}\cdot \rho^2\, d\rho\right)\cdot \int_0^{2\pi}\left(\int_0^{\pi} \sin\phi\, d\phi\right) d\theta$$

$u = \rho^3$

$$= \left(\int_0^1 e^{\rho^3}\cdot \rho^2\, d\rho\right)\cdot \left(\int_0^{\pi} \sin\phi\, d\phi\right)\cdot \int_0^{2\pi} 1\, d\theta$$

$$= \left[\frac{1}{3}e^{\rho^3}\right]_0^1 \cdot \Big[-\cos\phi\Big]_0^{\pi} \cdot \Big[\theta\Big]_0^{2\pi}$$

$$= \left(\frac{1}{3}e^1 - \frac{1}{3}e^0\right)\cdot (1-(-1))\cdot (2\pi - 0)$$

$$= \frac{1}{3}(e-1)\cdot 2\cdot 2\pi = \frac{4\pi}{3}(e-1).$$